Sheet: 1_inputs

Input	Value	Unit	Source	
Positive current collector thickness	8e-06	m	bridge p_r6_f2	
Negative current collector thickness	6e-06	m	bridge p_r6_f2	
Separator thickness	8e-06	m	bridge p_r6_f2	
Positive electrode thickness	9.828e-05	m	bridge p_r6_f2	
Negative electrode thickness	0.00012184	m	bridge p_r6_f2	
Nominal cell capacity	6.5	Ah	bridge p_r6_f2	
Electrode area	0.1027	m2	geometry block	
Negative / positive porosity	0.45 / 0.43	-	bridge p_r6_f2	
Separator porosity	0.47	-	Chen2020 set	
Positive electrode density	3262	kg/m3	Chen2020 set	
Negative electrode density	1657	kg/m3	Chen2020 set	
Separator density	397	kg/m3	Chen2020 set	
Al / Cu collector density	2700 / 8960	kg/m3	std Al/Cu	
Electrolyte conductivity	1.5	S/m	bridge p_r6_f2 ESTIMATE	
Cation transference number	0.6	-	bridge p_r6_f2 ESTIMATE	
Electrolyte diffusivity	5e-10	m2/s	bridge p_r6_f2 ESTIMATE	
Negative particle radius	3e-06	m	bridge p_r6_f2	
Total heat transfer coefficient	70	W/m2/K	bridge p_r6_f2	
Electrolyte fill density (literature)	1.2	g/cm3	LITERATURE ESTIMATE	
c_max neg / pos	33133 / 63104	mol/m3	Chen2020 set	
Active-volume fraction neg / pos	0.75 / 0.665	-	Chen2020 set	
Initial conc neg / pos	29866 / 17038	mol/m3	Chen2020 set	

Sheet: 2_mass

Layer	kg/m2	mass (g)	Share of contract mass (%)	
Positive electrode	0.182736	18.767	0.05	
Negative electrode	0.111039	11.4037	0.03	
Positive cc (Al 8um)	0.0216	2.2183	0.01	
Negative cc (Cu 6um)	0.05376	5.5212	0.01	
Separator (8um dry)	0.001683	0.1729	0	
TOTAL (contract caliber)	0.370818	38083	100	
Electrolyte fill (pore volume)	10.357	12.429	-	cm3; 1.2 g/cm3 ESTIMATE
TOTAL (BOM caliber)		50.512	-	~502 Wh/kg

Sheet: 3_energy

Quantity	Value	Unit	Source	
1C DFN discharge capacity	7.171	Ah	r7_final_f2_energy_dfn.json	
1C DFN discharge energy	25.375	Wh	same	
Contract mass	0.038083	kg	same (stack layers)	
Energy density (mass)	666.3	Wh/kg	E / m = 666.298 (full precision)	
Stack volume	2.4866e-05	m3	thickness x area	
Energy density (volume)	1020.5	Wh/L	E / V	
BOM-caliber mass incl electrolyte	0.0505	kg	bom sheet	
BOM-caliber ED	502.4	Wh/kg	honest packaging reality	
Midpoint voltage	3.78	V	r7 energy json	
DC resistance (mid-SOC)	2538	mohm	r7 energy json	
Specific power	42997	W/kg	r7 energy json	

Sheet: 4_stack_process

Process quantity	Formula	Value	Unit	
Positive areal loading	rho_pos x (1-por_pos) x L_pos	182.7	g/m2	
Negative areal loading	rho_neg x (1-por_neg) x L_neg	111	g/m2	
Positive compaction	rho x (1-por) / 1000	1.859	g/cm3	
Negative compaction	rho x (1-por) / 1000	0.911	g/cm3	
Electrolyte fill volume	Sum(t_i x por_i) x area	10.36	cm3	
Electrolyte fill mass	volume x 1.2 g/cm3	12.43	g ESTIMATE	
Anode stoich at full charge x0	c_init / c_max	0.9014	-	
Cathode stoich at initial x0	c_init / c_max	0.27	-	
Anode stoich after 1C discharge	x0 - dx	0.041	-	
Cathode stoich after 1C discharge	x0 + dx	0.902	-	
Anode areal capacity per dx	c_max x eam x L x F / 3600	81.15	Ah/m2	
Cathode areal capacity per dx	c_max x eam x L x F / 3600	110.54	Ah/m2	
Utilized areal capacity	Q_1C / A	69.82	Ah/m2	
N/P as-designed	neg Q to full-charge stoich / pos utilized	1.05	-	
Anode saturation headroom > full charge	(1 - x0_neg)/x0_neg	11.5	%	
1C max temperature	1C discharge DFN	300.42	K	
4C max temperature	4C charge DFN lumped, h=70	330.98	K	
4C anode potential min	4C charge DFN	0.0506	V	

Sheet: 5_verification

Contract criterion	Threshold	Measured	Verdict	Evidence
energy_density_wh_kg	>= 500.94	666.2985	PASS	r7_final_f2_energy_dfn.json / log-evaluate round 7
T_max_K	<= 333.15	330.984	PASS	r7_final_f2_4c45.json
plated	false	False (ap_min +0.0506 V)	PASS	r7_final_f2_4c45.json anode_potential_v